United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Mid-Term Exam Trimester: Summer 2025

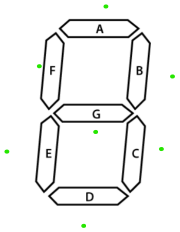
Course Code: CSE 4325 Course Title: Microprocessors and Microcontrollers

Total Marks: 30 Duration: 1 hour 30 minute(s)

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

Question 1: Answer all the questions. (6 Marks)

Transfer of bus control from processor to device takes **200 ns**. Transfer of bus control from device to processor takes **300 ns**.

a.	One of the I/O devices employs DMA in **cycle stealing mode** and takes **148,456,500 ns** to transfer a total **3648 Bytes** of data. The I/O device has a data transfer rate of **1440 KB/minute**. How many bytes are transferred at a time in this mode?	**[3]**
b.	In a Microcontroller, a seven segment display (like the image below) is connected to **PORTC**. **The A, B, C, D, E, F and G pins are connected to PC2, PC1, PC5, PC7, PC4, PC0 and PC6** respectively**. PC3 is not connected to anything.** If you want to display the digit **"6"**, what should be the value of the register **DDRC and why?** 	**[2+1]**

Question 2: Answer all the questions. (6 Marks)

a.	**AX = 3A53 H, BX = 4D31 H, CX = 2BA1 H,** Determine the values of the **PF and OF** after executing each of the following instructions sequentially. **Provide a brief explanation for your answer**. **i. ADD AX, BX** **ii. SUB AX, CX**	**[4]**
b.	What is meant by **multiplexed pins**? Give an example of such pins with reference to the 8086 microprocessor.	**[2]**

Question 3: Answer all the questions. (6 Marks)

a.	**CS = 1A32 H, DS = 6D2A H, SS = 7CA2 H** **SP = 3344 H, IP = 2211 H, SI = 11AA H** I) To access **the physical address** in the **Stack Segment**, what should be the value of the **segment register** if the **offset register** holds the value of **B294H**.	**[3]**

	II) To access **the physical address** in the **Data Segment**, what should be the value of the **offset register** if the **segment register** holds the value of **2B41H**. **Justify whether your answer is valid or not.**	
b.	A DRAM has **a data bus and an address bus** that are **twice** the width of the **data bus and address bus** of an SRAM. The capacities of the **SRAM and DRAM** are **8 KB and 64 MB, respectively**. Determine the width of the **address bus** of the **SRAM** and the width of the **data bus** of the **DRAM**.	**[3]**
Question 4: Answer all the questions. (6 Marks)		
a.	You need to design a system where a microprocessor with a **20-bit address bus** and **16-bit data bus** is interfaced to a **10 KB RAM** system using **full decoding**, starting from address **ACC00H**. Each RAM chip has a **10-bit address bus** and a **16-bit data bus**. **Draw a block diagram showing the microprocessor, RAM chips, and full decoding logic**. Ensure the first RAM **starts at ACC00H**. **Provide the 3rd, 5th, and 7th *last addresses* of the memory space occupied by the 2nd and 4th RAM chips.**	**[4+2]**
Question 5: Answer all the questions. (6 Marks)		
a.	If the size of the interrupt vector table is **2Kbyte** in an 8086 microprocessor, what will be **the address** of the **lower byte of CS** for the **last interrupt number**?	**[3]**
b.	An 8086 processor is executing an instruction at **physical address 1C203H**. **IP = 2003H**, **Flag Register = F202H**. **Describe the exact sequence of events** the 8086 performs when it encounters a software interrupt based on the given values and **show** the **stack contents**.	**[3]**